



KISANKAVACH

HARDWARE INFRASTRUCTURE SPECIFICATION & PROCUREMENT GUIDE

Comprehensive Technical Sizing, UIDAI L1 Compliance & Official GeM Procurement Blueprint

Project Author & Architect:	Sarath Babu Rayaprolu
Document Reference:	KK-HW-PROC-2026-001 (Version 1.0)
Deployment Geography:	Karnataka (10 Pilot Districts Expanding to All 31 Districts)
Target Implementers:	State Level Bankers' Committee (SLBC), Dept. of Agriculture, Grama One / CSCs
Statutory Mandates:	UIDAI L1 Biometric Mandate, STQC Certified, BIS Registered, DPDP Act 2023
Procurement Portals:	Government e-Marketplace (GeM) & Authorized OEM Direct Distribution
Classification:	Commercial-in-Confidence Official Technical Specification Memorandum

UIDAI L1 BIOMETRICS 100% Cryptographic Zero Replay / Spoof Proof	ASSISTED KIOSK FLEET 6,000+ Centers Grama One & Mandi Kiosks	BANK BRANCH DESKS 1,850+ Branches Maker-Checker Duplex Scanners
COMMAND CENTER WALL 31 Districts Collectorate Real-Time LFDs	PRIMARY SOURCING GeM Portal Govt e-Marketplace Verified	WARRANTY STANDARD 3 Yr On-Site + AMC Mission-Critical Field SLA

Executive Hardware Mandate: KisanKavach bridges the last-mile physical-to-digital divide across Karnataka's 8.2 million farmers and 1,850+ rural bank branches. Delivering 6-day loan sanctions requires rugged, tamper-proof hardware in rural taluks combined with high-throughput document scanners at bank desks. In strict compliance with UIDAI regulations, all biometric capture hardware specified herein utilizes Level 1 (L1) cryptographic key storage, ensuring biometric data is encrypted directly on-chip before leaving the sensor. Every specified device is fully procurable via the **Government e-Marketplace (GeM)** or through empaneled national distributors.

Document Prepared & Architected by: Sarath Babu Rayaprolu

1. Compliance Mandates & Engineering Standards

Deploying field biometric hardware across Karnataka's agricultural belt demands adherence to rigid national security standards:

STANDARD / REGULATION	STATUTORY AUTHORITY	MANDATORY SPECIFICATION FOR KISANKAVACH
UIDAI L1 Biometric Mandate	UIDAI (Aadhaar)	Must use Level 1 (L1) certified sensors where biometric encryption occurs within the crypto-boundary of the device. Legacy L0 devices are strictly rejected.
STQC Certification	MeitY / STQC	Mandatory Standardisation Testing and Quality Certification ensuring optical quality, false acceptance rate (FAR < 0.001%), and false rejection rate (FRR < 0.1%).
BIS Registration	Bureau of Indian Standards	Compliance with IS 13252 (Part 1) for information technology electrical safety under compulsory registration scheme (CRS).
Environmental Ruggedness	Field Standard	IP54 / IP65 dust and water splash resistance, drop-tested from 1.2 meters, operational temperature tolerance up to 50 deg C for rural field camps.

2. Category A: Field & Assisted Kiosk Fleet (Grama One / CSCs)

Deployed at Grama One centers, APMC Mandis, Primary Agricultural Credit Societies (PACS), and field verification officers:

DEVICE / COMPONENT	RECOMMENDED MODEL	KEY TECHNICAL SPECIFICATIONS	UNIT COST (INR)
A1. UIDAI L1 Fingerprint Scanner	Mantra MFS110 L1 (Alt: Morpho MSO 1300 E3 L1)	Optical sensor (500 DPI), hardware cryptoprocessor, on-chip AES-256 encryption, Android/Windows USB-C & micro-USB OTG, STQC & UIDAI L1 certified.	Rs. 3,200 - Rs. 3,850
A2. Dual-Eye Iris Scanner (For elderly/worn fingerprints)	Mantra MIS100V2 L1 (Alt: IriShield MK 2120U)	Spatial resolution > 60% @ 4.0 Lp/mm, pixel resolution 640x480, operating distance 10-15cm, STQC certified, dual LED infrared illumination.	Rs. 4,500 - Rs. 5,200
A3. All-in-One Rugged Bio-POS Tablet	Evolute Falcon 4G (Alt: BioRugged 7-Inch)	7.0" touch display, integrated L1 fingerprint sensor, 2" thermal printer, 4G Dual SIM, GPS/NavIC, 6000mAh battery, IP54 rugged casing.	Rs. 18,500 - Rs. 22,000
A4. Bluetooth Thermal Receipt Printer	NGX N-POS 2" (Alt: TVS RP 3200)	2-inch thermal printer (58mm), 203 DPI, Bluetooth 4.0 + USB, prints bilingual Kannada/English transaction receipts & KCC acknowledgment slips.	Rs. 2,400 - Rs. 3,100
A5. Solar Field Battery Pack	Ambrane / Mi 20,000mAh (Alt: EcoFlow 100W Kit)	20,000mAh rugged battery bank with 65W Power Delivery, dual USB-C output, solar panel charging input for off-grid taluk camp operations.	Rs. 1,800 - Rs. 2,400

3. Category B: Bank Branch Underwriting Desks (Maker-Checker)

Installed at 1,850+ participating commercial, RRB, and cooperative bank branches across Karnataka:

COMPONENT	RECOMMENDED MODEL	SPECIFICATIONS & OPERATIONAL PURPOSE	UNIT COST (INR)
B1. High-Speed Duplex Document Scanner	Epson WorkForce DS-530 II (Alt: Fujitsu fi-8170)	Sheet-fed duplex scanner, 35 ppm / 70 ipm, 50-sheet ADF, ultrasonic double-feed detection. Scans multi-page legacy land deeds, Pahani extracts, and mortgage charge memos.	Rs. 26,000 - Rs. 32,000
B2. FIPS Cryptographic USB Tokens	ePass2003 / ProxKey (Watchdata / Hypersecu)	FIPS 140-2 Level 3 certified USB cryptographic token. Stores Class 3 digital signature certificates (DSC) for Branch Managers to digitally sign sanction letters & lien memos.	Rs. 850 - Rs. 1,200
B3. Dual-Screen Underwriter Workstation	Dell OptiPlex 7010 / HP ProDesk 400 + Dual 24" FHD	Intel Core i5 (13th Gen), 16GB DDR5, 512GB NVMe SSD, dual 24-inch IPS monitors. Enables underwriters to view Bhoomi cadastral maps on Screen 1 and bank loan file on Screen 2.	Rs. 58,000 - Rs. 68,000

4. Category C: State Command Center & SLBC Oversight Hub

Deployed at the State SLBC Secretariat (Canara Bank HO, Bengaluru) and District Collectorates:

COMPONENT	RECOMMENDED MODEL	SPECIFICATIONS & OPERATIONAL PURPOSE	UNIT COST (INR)
C1. Command Center Video Wall LFD	Samsung 55" Video Wall (VH55R-R) (Alt: LG 55SVH7F)	55-inch ultra-narrow bezel (0.88mm), 700 nits brightness, 24/7 non-glare panel, daisy-chain DP 1.2. Displays live taluk heatmaps and SLA breach alerts.	Rs. 1,45,000 / panel
C2. Multi-Display GIS Controller	HP Z4 G5 Workstation (Alt: Dell Precision 3660)	Intel Xeon W-2400, 64GB ECC RAM, NVIDIA RTX A4000 (16GB), drives 4x 4K video wall outputs simultaneously for geospatial district drilldowns.	Rs. 1,85,000 - Rs. 2,15,000
C3. SLBC Virtual Meeting Endpoint	Logitech Rally Plus System (Alt: Poly Studio X50)	Ultra-HD 4K PTZ camera, dual beamforming speakerphones, automatic speaker framing for quarterly SLBC and district LDM coordination reviews.	Rs. 1,65,000 - Rs. 1,95,000

5. Category D: Data Center, Bank Gateway & Network Enclave

Required for State Data Centre (SDC) rack deployments and dedicated bank leased-line terminations:

COMPONENT	RECOMMENDED MODEL	SPECIFICATIONS & OPERATIONAL PURPOSE	UNIT COST (INR)
D1. Enterprise Next-Gen Firewall / VPN	Fortinet FortiGate 100F (Alt: Cisco Firepower 1150)	10 Gbps firewall throughput, 1 Gbps IPsec VPN throughput, redundant power supply, terminates point-to-point IPsec tunnels to Finacle/BaNCS CBS centers.	Rs. 2,85,000 - Rs. 3,40,000
D2. Hardware Security Module (HSM)	Thales Luna HSM A790 (Alt: Entrust nShield Connect)	FIPS 140-2 Level 3 certified network HSM. Cryptographic key isolation for Aadhaar Data Vault (ADV), database column encryption, and DPDP consent tokens.	Rs. 8,50,000 - Rs. 11,00,000

6. Official Procurement Sources & GeM Portal Workflow

For government entities (Dept. of Agriculture, District Collectorates) and Public Sector Banks (Canara Bank, SBI, KVGB), public procurement must strictly flow through the **Government e-Marketplace (GeM)** or authorized OEM national distributors under General Financial Rules (GFR) 2017.

ITEM CODE & CATEGORY	GeM PORTAL CATEGORY NAME	AUTHORIZED OEM / NATIONAL DISTRIBUTORS	PROCUREMENT METHOD
A1 & A2 Biometric Fingerprint & Iris Scanners	Biometric Attendance / Access Control & Aadhaar Identification Devices (L1 Certified)	• Mantra Softech India Pvt. Ltd. (Bengaluru / Ahmedabad) • IDEMIA (Morpho) India • Savex Technologies (National Distributor) • GeM Direct: Search 'MFS110 L1' or 'MIS100V2'	GeM Direct Purchase (< Rs. 5L) or L1 Custom Bid / PAC (Proprietary Article Certificate)
A3 & A4 Bio-POS Tablets & Bluetooth Printers	Point of Sale (POS) Terminals / Portable Micro-ATM Machines	• Evolute Devices Pvt. Ltd. (Mumbai / Bengaluru) • NGX Technologies Pvt. Ltd. (Bengaluru) • Ingram Micro India Pvt. Ltd. • GeM Product ID: Evolute Falcon POS	GeM e-Bidding with Custom Specification BoQ (Bill of Quantities)
B1 & B2 Duplex Document Scanners & DSC Tokens	Document Scanners (ADF Sheetfed) & Cryptographic USB Hardware Tokens	• Epson India Pvt. Ltd. (Bengaluru) • Redington India Ltd. (National Distributor) • e-Mudhra / SafeScript / VSign (DSC Tokens) • GeM Search: 'Epson DS-530 II' / 'ePass2003'	GeM Direct Buy / Rate Contract across Bank Regional Offices
B3 & C2 Dual-Screen Desks & GIS Workstations	Desktop Computers / Computer Monitors / High Performance Workstations	• Dell Technologies India • HP Enterprise India Pvt. Ltd. • Redington / Savex Technologies • GeM Product ID: OptiPlex 7010 / HP Z4 G5	GeM Forward Auction / OEM Bunch Bidding for volume discounts
C1 & C3 Video Wall LFDs & Video Conferencing	Large Format Displays (LFD) / Video Conferencing System	• Samsung India Electronics • Logitech Electronics India • Actis Technologies / Vega Global (AV SI Partners) • GeM Search: 'Samsung VH55R' / 'Rally Plus'	State e-Governance / CeG empaneled AV Turnkey tender
D1 & D2 Firewalls & Hardware Security Modules	Unified Threat Management (UTM) / Hardware Security Modules (HSM)	• Fortinet Technologies India • Thales India Pvt. Ltd. (Noida / Bengaluru) • Wipro / TCS / ITI Limited (System Integrators)	Bank IT Procurement / MeitY Empaneled Data Centre SI Contract

Important GeM Sourcing Tip: When raising bids on the GeM portal for Category A (Biometrics), procuring nodal officers must explicitly mandate **'UIDAI Level 1 (L1) Certificate Copy from STQC'** as a mandatory technical qualification criterion. Bids lacking this requirement often receive non-compliant legacy L0 devices which will be rejected upon connection to the UIDAI Aadhaar verification server.

7. Phase-Wise Hardware Procurement Budget & Quantities

Estimated procurement quantities and capital outlay structured across the four strategic rollout horizons:

ROLLOUT PHASE	GEOGRAPHIC & BRANCH SCOPE	KEY HARDWARE QUANTITIES	EST. BUDGET (INR)
PHASE 1: PILOT LAUNCH Months 1 – 6 (2026)	10 Pilot Districts • 350 Bank Branches • 1,200 Grama One / CSCs	• 1,200 Mantra MFS110 L1 Fingerprint Scanners • 150 Mantra MIS100V2 Iris Scanners • 350 Epson DS-530 II Duplex Scanners • 350 Workstations & DSC Tokens • 1 Pilot Command Center Display Unit	Rs. 3.45 Crore (\$415K USD)
PHASE 2: STATEWIDE KA Months 7 – 18 (2027)	All 31 Karnataka Districts • 1,850 Bank Branches • 6,000 Grama One / CSCs • 31 Collectorate Desks	• 4,800 Additional L1 Fingerprint Scanners • 600 Additional Iris Scanners • 1,500 Additional Duplex Scanners • 1,500 Additional Underwriter Workstations • 31 District Command Center LFD Displays • Redundant Fortinet Firewalls & HSM Enclave	Rs. 14.80 Crore (\$1.78M USD)
PHASE 3: REGIONAL SCALE Months 19 – 36 (2028-29)	Karnataka + AP, Telangana, TN • 4,200 Bank Branches • 15,000 Rural CSC Kiosks	• 9,000 L1 Biometric Scanners (PACS & Kiosks) • 2,350 Duplex Scanners for Regional Branches • 1,500 Rugged Evolute Bio-POS Tablets • State Data Centre High-Throughput HSM Expansion	Rs. 28.50 Crore (\$3.43M USD)
PHASE 4: PAN-INDIA SCALE Months 37 – 60 (2029-31)	7 Sovereign States • 16,000 Bank Branches • 50,000 Rural Touchpoints	• 35,000 L1 Biometric Scanners • 11,800 Bank Branch Underwriter Bundles • 8,000 Bio-POS Mobile Field Terminals • National AgriStack Sovereign Hardware Enclave	Rs. 92.00 Crore (\$11.08M USD)

8. Warranty, Spares & Field Maintenance Protocol

To prevent disruptions in rural credit delivery during the peak agricultural sowing window (Kharif & Rabi), the procurement RFP must mandate the following Service Level Agreement (SLA):

SLA PARAMETER	MANDATORY CONTRACTUAL REQUIREMENT
Warranty Period	Minimum 3-Year Comprehensive On-Site OEM Warranty covering all parts, sensor optical glass, and labor.
Buffer Spares in Taluks	5% operational spare inventory maintained at each District Lead Bank Office (LDM) for immediate swap within 24 hours.
Mean Time to Replace (MTTR)	< 24 hours in District Headquarters; < 48 hours in rural Taluk / Hobli centers.
Annual Maintenance Contract (AMC)	Years 4 and 5 covered under fixed-rate AMC capped at 8% of original equipment capital cost.

Document Author & Lead Architect: Sarath Babu Rayaprolu
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